

When ChatGPT Gives Incorrect Answers: The Impact of Inaccurate Information by Generative AI on Tourism Decision-Making

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Abstract

This study investigates how inaccurate information provided by ChatGPT impacts travelers' acceptance of recommendations. Six experiments were conducted based on the accessibility-diagnostics framework. These examined the moderating role of the prominence and type of incorrect information and their effects on decision-making. The results show that participants perceived more accuracy and trustworthiness, leading to stronger intentions to visit when incorrect information was absent. However, there was a decline in their intentions to visit when incorrect information was present and more prominent or in the same domain. This effect diminished when multiple domains were involved or when participants were focused on the initial task. The research highlights that both the prominence and type of incorrect information are boundary conditions and provides insights into AI applications in tourism. Furthermore, it offers practical implications for online travel agencies in terms of user interface and user experience design planning.

Keywords

ChatGPT, chatbot, AI recommendations, *accessibility-diagnostics*, incorrect information, generative AI

Introduction

The acceleration of technological innovations has transformed travel planning and booking practices. With the increasing prevalence of chatbots and virtual assistants, travelers can acquire instant access to travel information and personalized recommendations. Artificial intelligence (AI) powered Chatbots have grown in popularity because they address travel-related queries with accurate, quick, and personalized responses (Melián-González et al., 2021; Pillai & Sivathanu, 2020). ChatGPT is one of the most efficient and powerful chatbots for information search and is a large language model trained by OpenAI. This medium has been increasingly adopted in the travel and tourism fields (Stokel-Walker & Van Noorden, 2023; van Dis et al., 2023). The number of ChatGPT users exceeded 1 million just five days after launch and grew to two billion within two months. However, ChatGPT has some fundamental weaknesses which accompany its many innovative features. Notably, ChatGPT sometimes responds

with incorrect information to users when it provides official explanations.

This paper explores the role of exposure to incorrect ChatGPT-generated information in travel information search settings. As the fastest-growing application that has

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been launched, ChatGPT is likely to disrupt many industries, including hospitality and tourism (Gursoy et al., 2023). The primary objective of this study is to investigate the impact of incorrect information from ChatGPT on acceptance by travelers when searching for travel information. Focusing on the potentially negative effects of such incorrect information, this research examines the underlying mechanisms and boundary conditions that prospectively moderate the negative effect, such as the prominence of incorrect information. We then examine how prior exposure to incorrect information can affect subsequent tasks and suggest the influence of moderating variables on the negative effect. We will conduct an empirical exploration of the aforementioned prediction across six experimental studies, seeking to understand the potential limitations of chatbots such as ChatGPT. We will also seek insights into the factors impacting user trust and reliance on such technologies for travel recommendations.

This paper offers several contributions to the literature on chatbots and travel information search. First, it sheds light on the potentially negative impacts of incorrect ChatGPT-generated information on traveler acceptance of its recommendations. By highlighting potential limitations in the provision of accurate travel information, this study offers insights for chatbot designers and developers, as well as an enhanced theoretical understanding of human interactions with chatbots and AI. Second, the authors examine the boundary conditions that potentially moderate the negative impacts of incorrect information. Specifically, the prominence of incorrect information is identified as a critical boundary condition. By identifying such conditions, the study provides a more refined understanding of when and how chatbots such as ChatGPT can provide the most effective travel recommendations. Finally, this study extends research on the impacts of incorrect information by examining how subsequent tasks are affected by prior exposure, thereby providing insights into the long-term effects of chatbot recommendations on user trust and reliance. Our findings can inform chatbot design and development strategies that enhance user trust and reliance in the long run.

Literature Review

Understanding Misinformation and Fake News

Wrong, false, or even fake information and news have become a part of our daily lives since the advent of social media through the Internet. However, the information may be untrue with the intent to mislead or harm others (i.e., disinformation) or without intent (i.e., misinformation) (Cheng & Chen, 2021; Mishra & Samu, 2021; Wisker, 2021). Where misinformation and fake news may violate people's beliefs and values, they are likely to check out and evaluate the credibility of the relevant information. The results may either demonstrate anger and hate

(Wisker, 2021) and distrust toward the target (Cheng & Chen, 2021) or reveal a willingness to receive and share fake news, depending on the relevance of the content (Mishra & Samu, 2021).

Consumers need to leverage self-efficacy when detecting misinformation as well as fake news or reviews. Prior studies have indicated that while younger and tech-savvy consumers tend to recognize fake news relatively easily (Borges-Tiago et al., 2020), such self-efficacy increases their distrust of the targeted social media, thereby decreasing their use intensity through channels such as Facebook (Cheng & Chen, 2021). Various factors affect consumer attitudes toward fake news in this context, such as cross-cultural differences, product types, religiosity, and presentation format.

Borges-Tiago et al. (2020) found different levels of susceptibility to fake news among online users in European countries. Furthermore, Chang et al. (2020) revealed how hedonic and utilitarian products affect the positive link of information quality (i.e., completeness and accuracy) on consumer attitudes toward seller postings in the Facebook second-hand marketplace. Moreover, if fake news is closely intertwined with personal moral codes, such as following a halal food regime, then religiosity may predict attitudes toward brand hate triggered by feelings of anger (Wisker, 2021). Interestingly, Di Domenico et al. (2021) extended the research on fake news by demonstrating how the presentation format of source primacy (vs. headline primacy) can reduce its spread, either by decreasing trust in the message or increasing the deceptive intent of the source. Di Domenico et al. (2021) also discovered that this effect is strengthened when there are weak (vs. strong) interpersonal ties between the sharer and receiver.

Although the strength of a brand may somewhat negate the detrimental impact of brand-related misinformation and fake news (Mishra & Samu, 2021), Obadă (2019) has shown how global brands, such as Pepsi, New Balance, and Starbucks, can be casualties of fake news. Consumers threatened to boycott the Pepsi company, resulting in a fall of around 4% in the stock price because of fake news associated with Donald J. Trump supporters in the 2016 US presidential election. Notably, fake news conveyed misinformation about COVID-19, including vaccines, health, and the original cause of the outbreak. COVID-19-related misinformation has undeniably had a significant influence on the services sector, including tourism. Since the availability of information is essential for prospective tourists, fake news can change the underlying dynamics of collecting and processing information (Williams et al., 2022).

The prior literature has consistently emphasized misinformation, regardless of whether it was intentional or unintentional (Chang et al., 2020; Cheng & Chen, 2021; Williams et al., 2022). It has been noted that consumers

are operating in a new information era in which it is essential to check facts for accuracy and trustworthiness, thereby impacting our behavioral attitudes and intentions toward the specific product or brand.

Misinformation and AI

As concern about the production and spread of misinformation has grown, various minimization strategies have been proposed (e.g., Pennycook et al., 2021). Deploying AI offers one means of fighting misinformation (Horne et al., 2020). We can use AIs to mitigate the harmful effects of misinformation, whether this involves assessing the veracity of claims or marking news stories with AI-based credibility indicators. The primary advantage of AI in detecting misinformation lies in its ability to recognize patterns in complex data. Thus, AI is perceived as superior to human agents, not solely in mathematically complex domains but also in fields like clinical health, medicine, psychology, or even hiring (see Glikson & Woolley, 2020 for a review).

It is important to note that AI is imperfect despite the benefits that it brings. While excelling in specific domains, AI is still prone to errors. For example, voice assistants, AI chatbots, and AI-based trading systems can potentially make mistakes when processing human commands that do not follow their learned patterns, resulting in incorrect responses. Additionally, the responses generated by AIs tend to follow specific patterns, which may lack originality and context awareness, thus making them vulnerable to errors. As a result, people generally acknowledge that AI-based technologies are imperfect and are susceptible to various errors. Prior research on consumer adoption of AI has focused on two areas: (1) how consumer attitudes change following AI errors (e.g., Dietvorst & Bharti, 2020) and (2) what procedures can mitigate the negative effects of AI errors (e.g., Mahmood et al., 2022).

Recent advances and refinements in generative AI have addressed many of the preceding reliability concerns. The latest generative AI, such as ChatGPT, no longer relies on repeating learned patterns but instead generates unique responses, even to the same questions. Along with its eloquence and human-like answer styles, it is apparent that in most cases, the credibility of ChatGPT recommendations and responses is established quickly, despite potential vulnerability to various errors. It is worth noting that the main ChatGPT webpage acknowledges explicitly that it “*may occasionally generate incorrect information.*” Additionally, ChatGPT’s extensive user base creates a favorable environment because it serves as social proof. This has been shown to increase people’s trust, even in the case of imperfect algorithms (Alexander et al., 2018).

ChatGPT has received considerable attention due to its potential to enhance the customer experience and provide personalized recommendations. It is essential to

understand the implications of using ChatGPT in the tourism context, particularly its impacts on consumer behaviors. Table 1 presents a compilation of recent publications focusing on the application of ChatGPT in hospitality and tourism, with the key findings highlighted. While most articles are conceptual and provide theoretical frameworks for understanding the potential impact of ChatGPT, some offer empirical insights for implementation. It should be acknowledged that no research has specifically addressed the impacts on consumer behaviors when ChatGPT makes mistakes during tourism recommendations. The present study aims to investigate such scenarios.

AI Recommendation in Travel

AI can provide valuable information through recommendations based on capturing complex patterns observable within large datasets (De Bruyn et al., 2020). Today, many online travel agencies (OTAs) have integrated AI and machine learning into their platforms to help consumers navigate the products that they seek. In fact, various travel and hospitality organizations have deployed ChatGPT plugins, notably Expedia, Kayak, QuadLabs, Del Taco restaurant chain, and Magpie, offering useful and personalized information to customers (Dwivedi et al., 2023). With the increased demand for more tailored and personalized travel experiences (Kotiloglu et al., 2017; Rodríguez et al., 2012), travel recommendation systems are being adopted to reduce information overload and assist trip planning decisions (Liao & Zheng, 2018). However, existing research on AI-based recommendations in the tourism context has mainly involved proposing improvements to the quality of the system (Rodríguez et al., 2012). Shi et al. (2021) have also observed the need to investigate why and how consumers adopted such systems.

Trust is another critical factor associated with traveler decisions to adopt an AI-based recommendation system (Shi et al., 2021). Trust is strongly related to persuasive social interactions. It involves the reliability of promises and statements to achieve the expected outcomes (Di Domenico et al., 2021; Touré-Tillery & McGill, 2015). Trust is particularly crucial in situations involving uncertainty and risk (Mcknight et al., 2011). McKnight et al. (2011) have argued the importance of distinguishing characteristics of trust for interpersonal and technology exchange. They proposed three dimensions of trust for technology, namely: functionality, helpfulness, and reliability. Tussyadiah et al. (2020, p.2) defined trust in the information technology context as “a user’s expectation that information technology artifacts such as websites, virtual peers, recommendation agents, automated online assistants, robots and the likes will fulfill the expected responsibilities.” Trust also influences traveler willingness

Table 1. Summary of ChatGPT Literature.

Article	Conceptual or empirical paper	Main suggestions or findings
Ali et al. (2023)	Empirical	Chat GPT's relevance, credibility, usefulness, and intelligence significantly impacts perceived trust, which has positive impact on behavioral intentions.
Carvalho and Ivanov (2023)	Conceptual	ChatGPT and similar models are expected to significantly influence various tourism processes, improving customer service efficiency and boosting productivity in operations.
S. Chen et al. (2023)	Conceptual	The differences in perceptions of ChatGPT between Eastern and Western cultures can be attributed to individualism and collectivism.
Demir and Demir (2023)	Empirical	ChatGPT enhances service individualization and internalizing information, thereby fostering a positive effect on co-creation of service value.
Dogru et al. (2023)	Empirical	This study suggests a comprehensive framework for explaining the impact of generative AI usage in various sectors of hospitality and tourism industry, potentially leading to both value co-creation and co-destruction.
Dwivedi et al. (2023)	Conceptual	ChatGPT provides the opportunity to create systems that enable students to pose and respond to increasingly complex, and possibly more beneficial, questions.
Fesenmaier and Wöber (2023)	Conceptual	Utilizing ChatGPT could greatly enhance the process of creating and delivering service experiences in terms of efficiency.
Gursoy et al. (2023)	Conceptual	ChatGPT can assist teachers and students, but it may pose risks such as diminished critical thinking and educational inequality. The tourism industry can utilize ChatGPT for personalize services and content creation.
Iskender (2023)	Conceptual	ChatGPT could transform tourism education and research by generating texts, necessitating a reassessment of university teaching methods and assessment strategies.
Ivanov and Soliman (2023)	Conceptual	ChatGPT can be used in experiential learning projects in sport management, such as planning and executing an event in a course.
Keiper (2023)	Empirical	ChatGPT provides the opportunity to create systems that enable students to pose and respond to increasingly complex, and possibly more beneficial, questions.
J. Kim et al. (2023)	Empirical	Demonstrated the positive responses for the large recommended options suggestion by ChatGPT in travel.
J. H. Kim et al. (2023b)	Empirical	Demonstrated the positive effect of usage experience and negative effect of ChatGPT's mistakes on intention to use the ChatGPT in travel plan.
Koc et al. (2023)	Empirical	The management response produced by ChatGPT-4 meets the criteria for being both efficient and effective.
Mich and Garigiano (2023)	Conceptual	ChatGPT's applications could significantly impact digital transformation, posing unique challenges for societies, businesses, and individuals.
Nautiyal et al. (2023)	Conceptual	This article explores key questions about using ChatGPT in tourism academia, such as truthfulness, potential for plagiarism, co-authorship status, and its role in promoting interdisciplinarity and equity.
Shin and Kang (2023)	Empirical	The insights gained from ChatGPT suggest that the interdisciplinary nature of tourism research significantly benefits other academic fields, challenging conventional views in the field.
Wong et al. (2023)	Conceptual	This article illustrates various ways in which ChatGPT can enhance tourist experiences during pre-trip, en-route, and post-trip stages.
This paper	Empirical	This research demonstrates the negative effect of the incorrect information from ChatGPT as well as the mediators and moderators.

to accept information or suggestions from a robot (Salem et al., 2015). It has been suggested that trust is an “eloquent indicator for daily and sustained interactions” in decision-making contexts (Gaudiello et al., 2016, p.634). In all, it can be postulated that providing reliable information is critical to building trust in accepting AI-based recommendations.

Recommendation Acceptance

Although AI permeates our everyday lives, it is still in its early development. Users are thus faced with considerable

uncertainty. For example, the early literature on mobile technologies revealed that users had increasing concerns about the risks when purchasing tourism products through mobile devices (Luo et al., 2010). Travelers using AI-based recommendations may have similar concerns. Furthermore, decision-making is complicated by the intangibility of tourism and the fact that travelers need greater access to information about tourism experiences before the actual trip (S. Park & Tussyadiah, 2017). Moreover, the unique nature of AI-based technologies means that the adoption process involves both cognitive and emotional perspectives (Gursoy et al., 2019). Consequently, it has been suggested that

travelers utilize cognitive and affective processing (Seegebarth et al., 2019). In a similar vein, the literature on AI technology adoption has highlighted the importance of considering both cognitive and affective factors (Fan et al., 2020; Lin et al., 2020).

In the tourism and hospitality context, scholars such as Lu et al. (2019) and Sun et al. (2019) have examined how travelers adopt AI technologies. Notably, existing scholars have expressed mixed feelings about AI technologies (Shi et al., 2021). For example, users have higher expectations of service quality and performance from AI. At the same time, users feel uncomfortable and isolated by the human-like features of AI (Murphy et al., 2019). In addition, service robot encounters were evaluated more favorably when human-like agents used literal language (Choi et al., 2019). Whereas the quality of the output was a significant determinant of whether or not to adopt recommendations from traditional agents, Lin et al. (2020) discovered that the adoption of AI-based recommendation systems is impacted by the characteristics of the agent—when engaging with the system, users will evaluate both the performance and required effort. Similarly, considering the heterogeneity of travel contexts and the novel nature of AI devices, scholars recommend investigating both agent characteristics and environmental clues (Lu et al., 2019).

Algorithm aversion is another significant phenomenon that merits consideration. When algorithms and humans make the same mistake, people tend to have a greater loss of confidence in the case of algorithms (Dietvorst et al., 2015). Furthermore, people are more accepting of suggestions from algorithms when interventions are simple (J. Kim et al., 2021). Castelo et al. (2019) also demonstrated that people tend to place less trust in algorithms for subjective than objective tasks. While existing studies reflect the malleable nature of algorithm aversion, the rapid progress of AI-based development has been continuously transforming how people perceive AI and its capabilities. This has led to reports about the opposite phenomenon of algorithm aversion, namely algorithm appreciation (Logg et al., 2019). It is anticipated that recent ChatGPT breakthroughs will disrupt existing perceptions of AI and willingness to adopt AI-based recommendations.

Reimagining the Promise of Technological Salvation

The modern era has been marked by a prevalent acceptance that technology can provide a panacea for societal challenges and will ensure continuous economic growth. However, blind faith in technological capacities can overlook prospectively negative consequences. Awareness of such consequences has prompted scholars to advocate for greater consciousness of technologies, thereby ensuring positive societal impacts.

C. Fuchs (2009), for example, has critiqued the prevailing narrative, which assumes that technology will lead

inexorably to progress and societal improvements. He questions the power dynamics embedded within technological systems and argues that limiting and democratizing technological developments can foster a more inclusive society where technology is shaped to serve the interests of humanity rather than vice versa. Similarly, Samerski (2018) revisits Illich's critique and emphasizes the need to reevaluate our relationships with technology. Rather than as an end in itself, technologies can provide a means of achieving human well-being and sustainable development.

Furthermore, it is essential to embrace the limitations of technological developments in order to ensure that their implementation is responsible and ethical. Limitations can serve as safeguards against unintended consequences, imbalances of power, and prospectively negative effects. By establishing boundaries on technological reach, it is possible to create space for critical reflection, ethical considerations, and a shift toward alternative models such as degrowth (Samerski, 2009). Such limitations can ensure an alignment between technology, societal needs, and values (Emslie & Watts, 2017).

Tourism scholars Tribe and Mkono (2017) introduced the concept of "e-lienation" to highlight the potentially negative consequences of unfettered technology adoption. They argue that excessive reliance on technology in the tourism sector can undermine authentic experiences and genuine human connections. Moreover, Werthner (2022) promotes a digital humanism view of e-tourism, placing human interests and well-being at the forefront. Such democratization of technological developments allows for the creation of technologies that align with human values and promote both social and environmental responsibility. In addition, Gretzel et al. (2020) called for a transformative approach in e-Tourism research by adopting a critical approach to existing technological and scientific paradigms.

It is timely to reevaluate the notion that an overreliance on unlimited technological developments leads inexorably to progress and societal well-being. As chatbots and generative AIs gain prominence, their potential must also be examined critically. While they offer the potential benefits of efficiency and convenience, they should be guided by ethical considerations. Furthermore, there is a need to redirect the potential of technologies toward normative thought and human development. This requires a conscious utilization of technology as a tool to enhance human potential, facilitate equitable access to resources, and support sustainable development goals.

Development of Hypotheses

Main Prediction and Mediation

Travel platforms and agents have adopted and used consumer reviews and recommendations extensively to assist information searching and purchasing experiences.

Although scholars and practitioners have sought to improve the accuracy of recommendation systems, there is an ongoing prevalence of biased and inferior recommendations (Banker & Khetani, 2019). Nevertheless, as acceptance of AI technologies has grown, consumers have demonstrated more independence when searching for destination information or services and making their own decisions without the need for support from travel intermediaries. The emergence of ChatGPT, which can generate human-like text, has increased the likelihood that consumers will deploy this method when seeking information. However, ChatGPT may provide incorrect information and/or inferior recommendations, as was the case with user reviews and recommendations on traditional travel intermediaries (Bushwick & Mukerjee, 2022). This prompts the question of how consumers will react to ChatGPT-provided information and recommendations.

Online travel agents provide consumers with user reviews and recommendations that may reduce perceived uncertainty and risk regarding purchasing decisions (Y. F. Chen, 2008). When consumers evaluate user reviews or recommendations, they base their judgments on the extent to which the information is accurate and credible (C. M. K. Cheung et al., 2008; D. H. Park et al., 2007) along with its usefulness, sufficiency, and relevance (C. M. K. Cheung et al., 2008; M. Y. Cheung et al., 2009; H. W. Kim et al., 2004; D. H. Park et al., 2007). For example, Sparks and Browning (2011) posit that user reviews on travel and hospitality online communities can shape consumer beliefs and trust toward the services offered by a firm. Prior researchers have shown that trust plays an important role in individuals' adoption of information. Trust can determine whether consumers follow the recommendations and advice of online retailers (McKnight et al., 2002). Consumers are more likely to consider trustworthy recommendations when making decisions (Filieri & McLeay, 2014; D. H. Park et al., 2007). However, consumers will be less likely to adopt unreliable recommendations since they want to avoid being deceived. Filieri et al. (2015) argued that the more consumers view recommendations or word-of-mouth as trustworthy, the more likely they will follow.

Similarly, we predict that consumers will be less likely to adopt ChatGPT-generated recommendations when they contain incorrect information. If consumers find that the information generated by ChatGPT is inaccurate and untrustworthy, they will be less willing to adopt the information and follow the recommendations. Thus, we propose the following hypotheses:

H1: *Incorrect information provided by ChatGPT could reduce users' reliance on ChatGPT for recommendations and their likelihood of visiting the recommended destinations.*

H2: *The perceived accuracy and trustworthiness of the information will mediate H1.*

The Moderating Role of the Prominence of Incorrect Information

We further investigate the prominence or salience of incorrect information. Previous scholars have suggested that the salience of information relates directly to its attractiveness (Dejica, 2009) and its relative importance or weight in decision-making (J. Kim et al., 2014). In other words, if incorrect information is salient, such as listing a relatively early (vs. later) position in multiple listings (e.g., locating 3rd [vs. 17th] position out of 20 listings), there is likely to be a stronger impact of the incorrect information on other judgments and decision-making. The prominence of incorrect information may be expected to have a large effect, especially in situations of AI-generated messaging, since consumers perceive machine errors more negatively than their human equivalents. For example, Dietvorst et al. (2015) suggested that people prefer to make use of humans (vs. algorithms) when they observe forecasting errors.

Based on the preceding observations, we may expect that the prominence of incorrect information will have a significant influence on acceptance of ChatGPT's recommendations and visit intentions for recommended places.

H3: *The prominence of incorrect information will moderate the impact of incorrect information on ChatGPT's recommendation acceptance and visit intention for recommended destinations.*

How Being Exposed to Incorrect Information Impacts on Subsequent Tasks

Since ChatGPT is readily accessible for various tasks, various reasons can be proposed for its deployment. In the following section, we focus primarily on how prior exposure to inaccurate or incorrect information impacts subsequent tasks.

For instance, imagine a traveler searching for information on destinations or attractions within an unfamiliar city. She/he may have previously experienced a mistake by ChatGPT in a similar or different domain. In this situation, we can expect that the negative effect on subsequent information-seeking of the ChatGPT mistake will be stronger when users have experienced an error within the same (vs. a different) domain. This prediction draws upon the *accessibility-diagnostics* framework (Herr et al., 1991; Menon et al., 1995), which assumes that information usage depends on the ease of accessibility to the information when making judgments and decisions, as well as the relevance of the information for the given task.

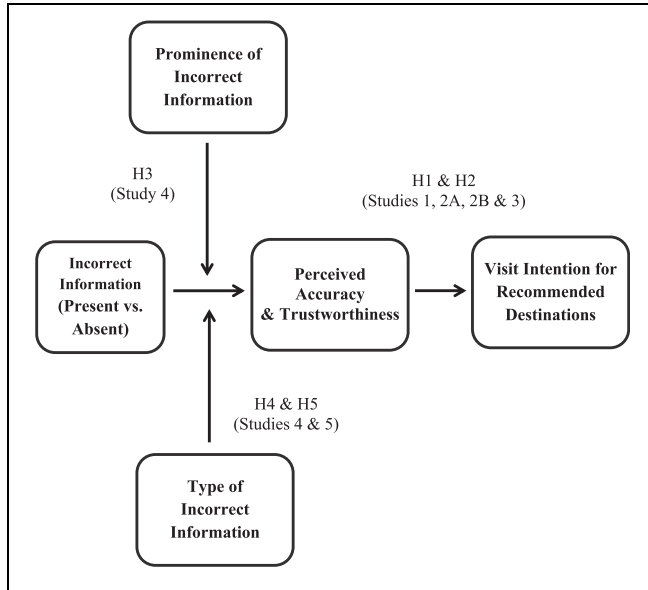


Figure 1. Theoretical framework and empirical studies.

The different domains of error in the previous scenario are conceptually related to the diagnosticity of incorrect information. It is expected that the impact of incorrect information will be negative when it is within the same domain, since it is highly diagnostic for a subsequent and similar task. On the other hand, when the incorrect information occupies a domain that is different from the subsequent task, the impact may be minimal since the incorrect information is not diagnostic for the subsequent task. This pattern is expected to be stronger when the incorrect information is highly salient, such as when judging whether the incorrect information is accurate.

Based on the preceding observations, we expect a significant moderating effect on the types of incorrect information. The following provides a formal explanation of the specific prediction:

H4: *The types of incorrect information will moderate the impact of incorrect information on reliance on ChatGPT. When incorrect information is related to similar tasks, a negative effect of ChatGPT's incorrect information is expected. In contrast, when incorrect information pertains to different types of tasks, the negative effect of ChatGPT's incorrect information will be reduced.*

In the following section, we mainly discuss the boundary conditions for the negative effect of incorrect ChatGPT information, especially within the same domain. We further suggest that the negative effects of incorrect information can be significantly reduced when ChatGPT users focus on other tasks instead of using the mistake to make assessments of accuracy.

Using the previous scenario as an example, if a ChatGPT user is focused on decision-making in the initial task with incorrect information, the negative effect of ChatGPT on a similar subsequent task will be significantly reduced. This argument is based on the accessibility components of the accessibility-diagnostics framework. When a consumer is highly focused on one aspect of a task, it is easy to neglect other information. One potential occurrence is *inattentional blindness* (Mack, 2003; Simons, 2000). It may be anticipated that the impact of previous exposure to mistakes in an earlier task will be low when users are occupied with other tasks.

Based on this logic, we suggest the following boundary conditions for the previous hypothesis (H4). We anticipate that when users are occupied with selecting one option in the initial task, the impact of incorrect information on subsequent decisions may be relatively low, even in the case of similar tasks.

H5: *The different focus on the initial task will significantly influence reliance on ChatGPT in the subsequent task. Specifically, when users are occupied with non-accuracy-related operations in the initial task, the negative effect of ChatGPT in the initial task on the subsequent task will be significantly reduced.*

The overall study framework is explained in Figure 1. The various empirical components were conducted in February 2023 (for Studies 1, 3, 4, & 5), a period when ChatGPT was receiving intense public attention, followed by a successive round in July 2023 (for Studies 2A & 2B). All materials used in the stimuli were based on ChatGPT version 3 (for studies 1, 3, 4, & 5) and 3.5 (for Studies 2A & 2B) without a paid membership. After conducting the first phase (Study 1), the sample size was pre-determined by utilizing the G*Power program (Faul et al., 2007). Based on the study 1 results, we established the criteria applicable to studies 2 to 5 (i.e., effect size $f = 0.25$ [medium], $\alpha = .05$, $1 - \beta = .95$) of 2×2 ANOVA (Analysis of Variance). Since the minimum sample size was determined to be 210, we tried to collect at least 210 respondents per study. The participant demographics are outlined in Table 2.

Study 1: Providing Evidence of the Negative Effects of Incorrect Information by ChatGPT

In Study 1, we aim to provide initial evidence of the negative effect of incorrect ChatGPT-provided information on the reliance that customers place on ChatGPT. Specifically, we anticipate that there will be lower visit intentions for the ChatGPT recommended destination when it contains incorrect information. For the purposes

Table 2. Profile of participants.

		Study 1 (n = 153) (%)	Study 2A (n = 222) (%)	Study 2B (n = 214) (%)	Study 3 (n = 298) (%)	Study 4 (n = 335) (%)	Study 5 (n = 325) (%)
Location		USA	USA	UK	USA	USA	USA
Gender	Male	51.6	50.5	43.5	43.6	45.4	45.5
	Female	48.4	49.5	56.5	56.4	54.3	53.2
Age	18–29	19.0	18.5	10.3	19.5	18.5	19.4
	30–39	36.6	33.3	25.8	31.2	35.5	33.8
	40–49	21.6	19.4	20.7	21.5	22.1	26.8
	50–59	13.7	14.9	21.6	13.8	16.1	11.1
	60 and above	9.2	14.0	21.7	14.1	7.8	8.9
Race	White/Caucasian	73.9	74.3	92.5	69.1	73.7	70.8
	African (American)	8.5	14.4	1.9	12.1	8.4	12.6
	Hispanic	6.5	2.7	0.0	7.0	7.2	4.0
	Asian	7.2	7.2	3.3	7.4	7.5	9.2
	Others	4.0	1.4	2.3	4.3	3.3	3.4
Education level	Did not complete high school	0.7	0.9	0.5	0.3	0.3	0.3
	High school graduate or some college	39.2	32.4	39.3	33.2	32.8	36.3
	College graduate (4 years)	41.8	36.0	40.2	44.0	42.7	42.8
	Postgraduate degree	18.3	30.6	20.1	22.5	24.2	20.6
Household income	Under \$30,000	21.6	13.5	26.2	18.1	17.6	18.2
	\$30,001–\$60,000	35.9	29.7	33.2	32.6	27.8	27.4
	\$60,001–\$90,000	16.3	23.0	24.3	19.8	22.7	23.1
	\$90,001–\$120,000	14.4	16.2	9.3	13.1	17.3	12.9
	\$120,001 and above	11.8	17.6	7.0	16.4	14.6	18.5

of the current study, New Zealand was chosen as the target travel destination.

Method: Participants, Procedure, and Measurement

There were 153 US participants in the study ($M_{\text{age}} = 40.38$, $SD = 12.23$; 48.4% female) were drawn from Amazon Mechanical Turk in exchange for a nominal payment. Participants were randomly assigned to one of two conditions (incorrect information: absent vs. present) with a between-subjects design.

First, participants were asked to imagine that they were planning a trip to the North Island of New Zealand and were seeking information on destinations/activities during their trip and that they turned to ChatGPT to identify recommendations. ChatGPT listed 20 destinations to visit in the North Island of New Zealand. As shown in Figure 2, the recommended places were provided as screenshots from an actual ChatGPT interaction. Those participants who were presented with the incorrect information condition received the following additional information: “After that, you found that ChatGPT provided totally wrong information. The 17th recommended destination, Lake Tekapo, is actually located in the South Island, not the North Island.” Participants in the incorrect information absent condition were exposed to the same list without the additional information (see below).¹

All participants were subsequently asked to evaluate their intentions to visit the recommended places (i.e., how much would you like to visit the recommended places)

with two items along a 7-point scale (i.e., 1 = not at all/very low, 7 = very much/very high). Participants were then asked to rate their perception of accuracy for the recommended places, with two items along a 7-point scale (i.e., 1 = not at all accurate/not at all objective, 7 = very accurate/very objective) as well as the perceived trustworthiness with two-items (i.e., 1 = not at all credible/not at all trustworthy, 7 = very credible/very trustworthy). Subsequently, participants were asked to evaluate the realism of this study on a 7-point scale (i.e., 1 = highly unrealistic, 7 = highly realistic).

Next, participants were asked to share their knowledge of and experience with ChatGPT (e.g., Have you heard of ChatGPT? Or Have you used ChatGPT?, yes or no) and provide demographic information.

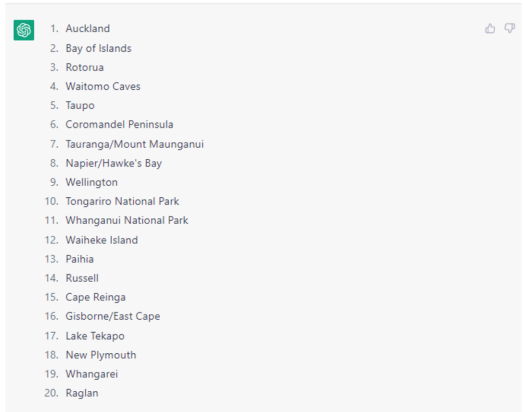
Results and Discussion

The perceived realism was relatively high in that the value was higher than the neutral point ($M = 5.29$, $SD = 1.44$ vs. “4,” $t(152) = 11.09$, $p < .001$).

For the main analysis, we conducted one-way ANOVA for three key measurements. First, for the intention to visit the recommended places, we found that the experimental factor had a significant effect. The results indicated that the intention to visit was higher when incorrect information is absent versus present ($M_{\text{incorrect information absent}} = 5.31$, $SD = 1.41$ vs. $M_{\text{incorrect information present}} = 4.05$, $SD = 1.71$; $F(1, 151) = 24.74$, $p < .001$, $\eta^2 = 0.141$), as shown in Figure 3.

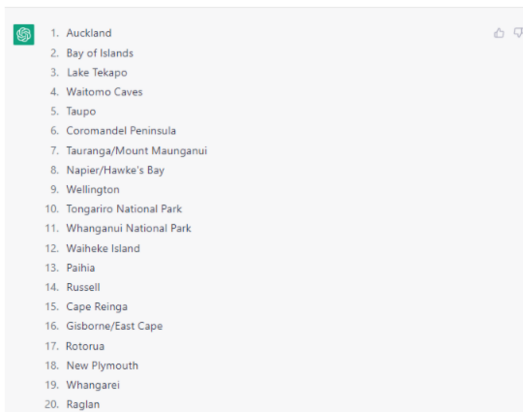
Studies 1 & 3: Incorrect information present condition (in 17th position & in 3rd position)

K List 20 visiting destinations around the North Island of New Zealand.



**** After that, you found that ChatGPT provided totally wrong information. The 17th recommended destination, Lake Tekapo, is actually located in the South Island, not the North Island.**

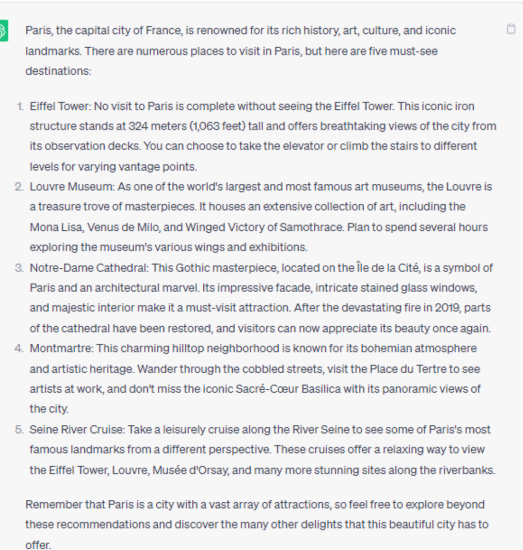
K List 20 visiting destinations around the North Island of New Zealand.



**** After that, you found that ChatGPT provided totally wrong information. The 3rd recommended destination, Lake Tekapo, is actually located in the South Island, not the North Island.**

Studies 2A & 2B: Incorrect information absent & present conditions

K Please recommend 5 places to visit in Paris, France



K Please recommend 5 places to visit in Paris, France

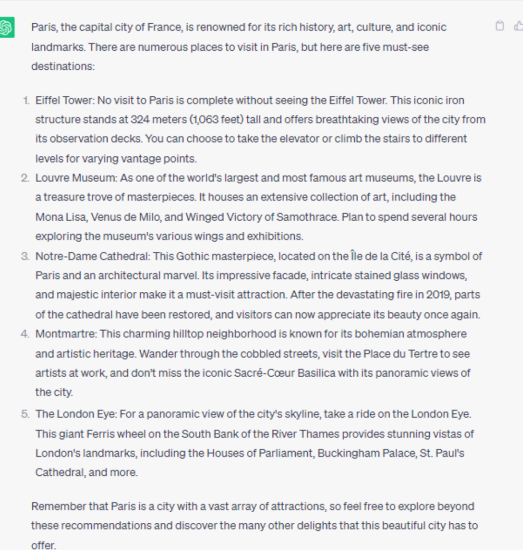


Figure 2. Stimuli of studies 1, 2, & 3.

Second, we found a significant effect for perceived accuracy. The results indicated that the perceived accuracy was higher when incorrect information is absent versus present ($M_{\text{incorrect information absent}} = 5.10$, $SD = 1.25$ vs. $M_{\text{incorrect information present}} = 4.07$, $SD = 1.65$; $F(1, 151) = 19.03$, $p < .001$, $\eta^2 = 0.112$).

We also found a similar significant effect for perceived trustworthiness ($M_{\text{incorrect information absent}} = 5.29$, $SD = 1.33$ vs. $M_{\text{incorrect information present}} = 3.95$, $SD = 1.72$; $F(1, 151) = 28.63$, $p < .001$, $\eta^2 = 0.159$).

Finally, we conducted a serial mediation (IV $\rightarrow$ perceived accuracy $\rightarrow$ perceived trustworthiness $\rightarrow$ visit intention) using Hayes (2017) macro with 5,000 bootstrapping. The results indicated that the indirect effect of serial mediation was significant (95% CI $[-0.605, -0.067]$), supporting H2. The detailed results are illustrated in Figure 4.

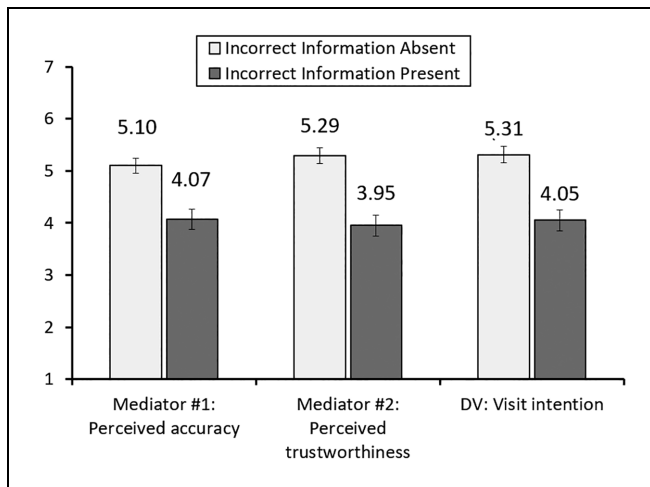


Figure 3. Results of Study 1.

*Error Bars represent the standard error.

Studies 2A and 2B: Extending the Validity of the Previous Study

Though we provided initial evidence of the negative effects of incorrect information provided by ChatGPT, we used extra wording as a reminder of erroneous information. In this study, we replicated the earlier investigative phase without this extra reminder. Furthermore, in order to extend the external validity, we conducted two studies deploying different online panels, participant source countries, and timing of the survey.

Method: Participants, Procedure, and Measurement

Drawn from CloudResearch, participants in Study 2A consisted of 222 US residents ($M_{\text{age}} = 41.92$, $SD = 13.70$; 49.5% female). Participants in Study 2B were drawn from a Prolific online panel and consisted of 214 UK participants ($M_{\text{age}} = 46.13$, $SD = 13.44$; 56.5% female) who received a nominal payment. Using a between-subjects design, participants were randomly assigned to one of two conditions (incorrect information: absent vs. present).

The overall procedures were similar, with few modifications. Participants were initially asked to imagine that they were planning a trip to Paris, France, and seeking information with ChatGPT. They were shown the five

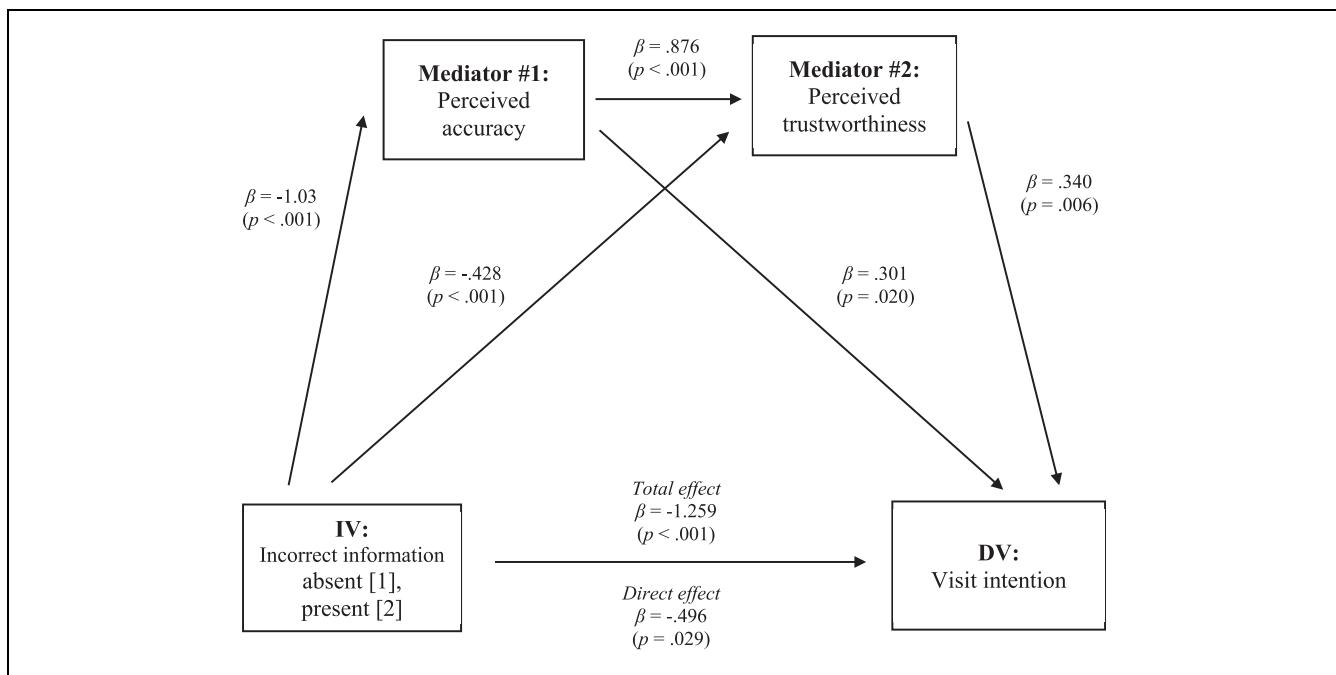


Figure 4. Serial Mediation Results of Study 1.

*Indirect effect.

- Incorrect information $\rightarrow$ Perceived accuracy $\rightarrow$ Visit intention, 95% CI $[-0.701, -0.015]$.
- Incorrect information $\rightarrow$ Perceived trustworthiness $\rightarrow$ Visit intention, 95% CI $[-0.337, -0.017]$.
- Incorrect information $\rightarrow$ Perceived accuracy $\rightarrow$ Perceived trustworthiness $\rightarrow$ Visit intention, 95% CI $[-0.605, -0.067]$.

places recommended by ChatGPT, including the Eiffel Tower and the Louvre Museum. As is shown in Figure 2, the experimental factor was manipulated with the information of the fifth recommended place. Participants in the incorrect information absent condition were given “Seine River Cruise,” whereas those in the incorrect information present condition were exposed to “The London Eye.” Using the same scale that was used in Study 1, participants were then asked to evaluate their intentions to visit the recommended places along with their perceived accuracy, trustworthiness, and realism of the scenario.

Next, participants were asked to check whether they noticed significant errors or mistakes in the recommended places as yes or no.

Results and Discussion of Study 2A

The perceived realism was relatively high in that the value was higher than the neutral point ($M = 5.93$, $SD = 1.14$ vs. “4,” $t(221) = 25.22$, $p < .001$). Regarding the recognition of errors in ChatGPT recommendations, participants recognized errors much more frequently when incorrect information was present versus absent ($M_{\text{incorrect information present}} = 38.7\%$ [43/111] vs. $M_{\text{incorrect information absent}} = 6.3\%$ [7/111], $\chi^2(1)$, $p < .001$).

First, we found a significant effect on the intention to visit the recommended places in that the intention to visit was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.87$, $SD = 1.10$ vs. $M_{\text{incorrect information present}} = 5.49$, $SD = 1.56$; $F(1, 220) = 4.38$, $p = .038$, $\eta^2 = 0.020$), as shown in Figure 5.

Second, we found a similar pattern for perceived accuracy in that the perceived accuracy was higher when incorrect information is absent versus present ($M_{\text{incorrect information absent}} = 5.73$, $SD = 0.92$ vs. $M_{\text{incorrect information present}} = 5.09$, $SD = 1.46$; $F(1, 220) = 14.75$, $p < .001$, $\eta^2 = 0.063$). We also found a similar significant effect for perceived trustworthiness ($M_{\text{incorrect information absent}} = 5.80$, $SD = 0.95$ vs. $M_{\text{incorrect information present}} = 5.18$, $SD = 1.57$; $F(1, 220) = 21.14$, $p < .001$, $\eta^2 = 0.054$).

Finally, we conducted a serial mediation (IV $\rightarrow$ perceived accuracy $\rightarrow$ perceived trustworthiness $\rightarrow$ visit intention) using Hayes’s (2017) macro with 5,000 bootstrapping. The results indicated that the indirect effect of serial mediation was significant (95% CI $[-0.618, -0.143]$), supporting H2.

Results and Discussion of Study 2B

The perceived realism was relatively high in that the value was higher than the neutral point ($M = 5.44$, $SD = 1.42$ vs. “4,” $t(213) = 14.88$, $p < .001$). Regarding the recognition of errors in ChatGPT recommendations, participants recognized errors much more frequently when incorrect information was present versus absent ($M_{\text{incorrect information present}} = 82.7\%$ [86/104] vs. $M_{\text{incorrect information absent}} = 4.5\%$ [5/110], $\chi^2(1)$, $p < .001$).

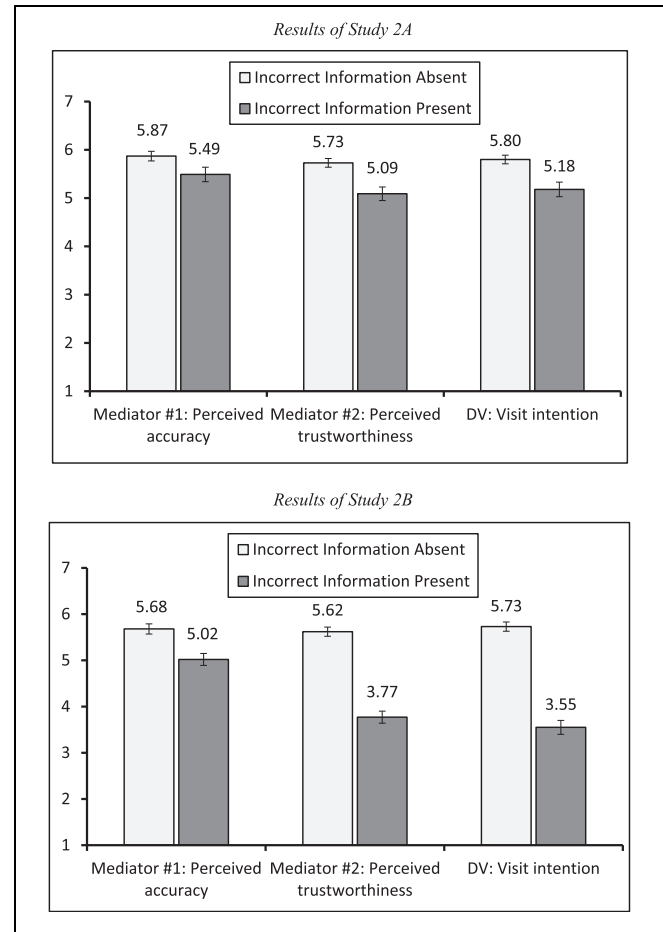


Figure 5. Results of studies 2A & 2B.

information present = 82.7% [86/104] vs. $M_{\text{incorrect information absent}} = 4.5\%$ [5/110], $\chi^2(1)$, $p < .001$).

First, we found a significant effect for the intention to visit the recommended places in that the intention to visit was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.68$, $SD = 1.16$ vs. $M_{\text{incorrect information present}} = 5.02$, $SD = 1.31$; $F(1, 212) = 15.02$, $p < .001$, $\eta^2 = 0.066$), as shown in Figure 5.

Second, we found a similar pattern for perceived accuracy in that the perceived accuracy was higher when incorrect information is absent versus present ($M_{\text{incorrect information absent}} = 5.62$, $SD = 1.02$ vs. $M_{\text{incorrect information present}} = 3.77$, $SD = 1.32$; $F(1, 212) = 132.72$, $p < .001$, $\eta^2 = 0.385$). We also found a similar significant effect for perceived trustworthiness ($M_{\text{incorrect information absent}} = 5.73$, $SD = 1.06$ vs. $M_{\text{incorrect information present}} = 3.55$, $SD = 1.56$; $F(1, 212) = 144.92$, $p < .001$, $\eta^2 = 0.406$).

Finally, the results of a serial mediation (IV $\rightarrow$ perceived accuracy $\rightarrow$ perceived trustworthiness $\rightarrow$ visit intention) indicated that the indirect effect of serial mediation was significant (95% CI $[-0.768, -0.157]$), supporting H2.

Study 3: Testing Moderating Role of Prominence of Incorrect Information (H3)

In Study 3, we investigated the moderating effect of the prominence of the incorrect information generated by ChatGPT. Specifically, when the incorrect information generated by ChatGPT is prominent (such as 3rd vs. 17th information out of 20), it is anticipated that exposure to the incorrect information will have a stronger negative effect.

Method: Participants, Procedure, and Measurement

The 298 study participants were from the USA ($M_{\text{age}} = 42.02$, $SD = 13.61$; 56.4% female) and were drawn from Amazon Mechanical Turk in exchange for a nominal payment. Participants were randomly assigned to one of four groups: 2 (incorrect information: absent vs. present) $\times$ 2 (prominence of incorrect information: low [17th incorrect information] vs. high [third incorrect information]) conditions with a between-subjects design.

Except for a few modifications, the general study procedure resembled Study 1, including the scenario of information seeking for future travel to New Zealand. The manipulation of incorrect information resembled the conduct of Study 1. To manipulate the prominence of incorrect information, the location of the incorrect information of “Lake Tekapo” was placed in either the 3rd or 17th out of 20 lists, as is demonstrated in Figure 2. All participants were then asked to indicate their visit intentions for the recommended destinations, following the scaling used in Study 1.

Results and Discussion

We conducted a 2×2 ANOVA to understand travelers' intentions to visit. The results indicated that the main effect of incorrect information was significant ($F(1, 294) = 24.43$, $p < .001$, $\eta^2 = 0.077$) in that the visit intention was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.08$, $SD = 1.51$ vs. $M_{\text{incorrect information present}} = 4.14$, $SD = 1.77$), replicating the result of Study 1 as well as supporting H1. However, the main effect of the prominence of incorrect information was not significant ($F(1, 294) = 0.60$, $p = .438$, $\eta^2 = 0.002$).

Importantly, as demonstrated in Figure 6, the interaction effect was significant ($F(1, 294) = 3.90$, $p = .049$, $\eta^2 = 0.013$). Planned contrast indicated that when the incorrect information was in 17th position, the visit intention was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 4.97$, $SD = 1.63$ vs. $M_{\text{incorrect information present}} = 4.40$, $SD = 1.67$, $F(1, 294) = 4.41$, $p = .037$, $\eta^2 = 0.015$). In addition, when the incorrect information was in a prominent position of third out of a list of 20, the visit intention was much higher when incorrect information was absent

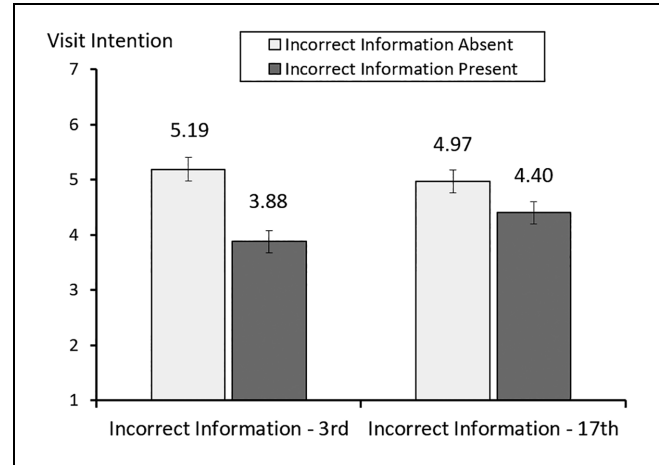


Figure 6. Results of Study 3.

*Error bars represent the standard error.

versus present ($M_{\text{incorrect information absent}} = 5.19$, $SD = 1.39$ vs. $M_{\text{incorrect information present}} = 3.88$, $SD = 1.84$, $F(1, 294) = 23.92$, $p < .001$, $\eta^2 = 0.075$). Analyzed differently, when incorrect information was absent, the visit intention was similar, regardless of the position of the incorrect information ($M_{\text{3rd incorrect information}} = 5.19$, $SD = 1.63$ vs. $M_{\text{17th incorrect information}} = 4.97$, $SD = 1.63$, $F(1, 294) = 0.71$, $p = .400$, $\eta^2 = 0.002$). However, when incorrect information was present, the visit intention was much lower when incorrect information was located in prominent (vs. less prominent) positions ($M_{\text{3rd incorrect information}} = 3.88$, $SD = 1.84$ vs. $M_{\text{17th incorrect information}} = 4.40$, $SD = 1.67$, $F(1, 294) = 3.81$, $p = .052$, $\eta^2 = 0.013$).

Study 4: Testing the Moderating Role of Type of Incorrect Information on Subsequent Decision-Making (H4)

The previous studies focused primarily on visit intentions to the recommended destinations when the recommendation itself contained or did not contain incorrect information. In Study 4, we focused on exposure to the previous recommendation tasks and the subsequent decision. We also considered the relationship between the initial and subsequent tasks: examples include a similar one (e.g., travel search task I & travel search task II) or a different one (e.g., music search task & travel search task). We anticipated that the negative effects of previous exposure to incorrect information would be higher when the two tasks are in the same domain. In contrast, the negative effect will likely have a significant reduction when the two tasks differ.

Method: Participants, Design, and Procedure

There were 335 study participants from the USA ($M_{\text{age}} = 40.49$, $SD = 11.99$; 54.3% female), drawn from

Same domains vs. different domains (with incorrect information present)

Information Searching with ChatGPT

Imagine that you asked **ChatGPT**, an AI-based online chatbot developed by OpenAI, to list **10 destinations** to visit around the **North Island of New Zealand**. ChatGPT responded with the following.

K List 10 visiting destinations around the North Island of New Zealand.

1. Auckland
2. Bay of Islands
3. Lake Tekapo
4. Waitomo Caves
5. Taupo
6. Coromandel Peninsula
7. Tauranga/Mount Maunganui
8. Napier/Hawke's Bay
9. Wellington
10. Tongariro National Park

**** After that, you found that ChatGPT provided totally wrong information. The 3rd recommended destination, Lake Tekapo, is actually located in the South Island, not the North Island.**

Information Searching with ChatGPT

Imagine that you asked **ChatGPT**, an AI-based online chatbot developed by OpenAI, to list **10 classical composers** who lived **before the year 1800**. ChatGPT responded with the following:

K List 10 famous classical composers who lived before the year 1800.

1. Johann Sebastian Bach
2. Wolfgang Amadeus Mozart
3. Robert Schumann
4. Antonio Vivaldi
5. George Frideric Handel
6. Johann Strauss II
7. Franz Joseph Haydn
8. Antonio Salieri
9. Claudio Monteverdi
10. Giuseppe Verdi

**** After that, you found that ChatGPT provided totally wrong information. The 3rd composer, Robert Schumann (1810-1856), actually lived after the year 1800.**

Main task

K Please recommend some destinations to visit during a traveler's trip to South Island, New Zealand

Certainly! Here are some recommended destinations for a traveler's trip to the South Island of New Zealand:

1. Milford Sound: A stunning fiord known for its dramatic landscapes and wildlife sightings.
2. Queenstown: A popular destination for adventure seekers, with activities such as bungee jumping, skydiving, and skiing.
3. Fox Glacier: A glacier located in Westland Tai Poutini National Park that is accessible for hiking and guided tours.
4. Abel Tasman National Park: Known for its beautiful coastline and hiking trails.
5. Mount Cook: The highest peak in New Zealand, surrounded by stunning alpine scenery and hiking trails.
6. Kaikoura: A coastal town famous for its whale watching tours and seafood.
7. Wanaka: A scenic town known for its outdoor activities, including hiking, skiing, and water sports.
8. Franz Josef Glacier: A glacier located in Westland Tai Poutini National Park that is accessible for hiking and guided tours.
9. Marlborough Sounds: A network of waterways, bays, and inlets known for its scenic beauty and outdoor activities.
10. Lake Tekapo: Known for its stunning turquoise waters, beautiful starry skies, and outdoor activities such as hiking, stargazing, and skiing.

Figure 7. Stimuli of study 4.

Amazon Mechanical Turk in exchange for a nominal payment. Participants were randomly assigned to one of two conditions (incorrect information: absent vs. present) $\times$ 2 (type of information: similar domains vs. different domains) with a between-subjects design.

All participants were informed about the multiple tasks in this study. The first task depended on how the information was manipulated. Participants in the similar domains group were asked to imagine that they had asked ChatGPT for a listing of 10 destinations for them to visit

around New Zealand's North Island. They were then shown the list (see Figure 7). Participants in the incorrect information present condition were also exposed to the mistaken listing of "Lake Tekapo" as a North Island destination, similar to previous Study 1, whereas those in the absent condition were not. Participants in a different domain condition were asked to imagine that they asked ChatGPT to list ten classical composers who lived prior to the year 1800. As is shown in Figure 7, participants in the incorrect information present condition were also

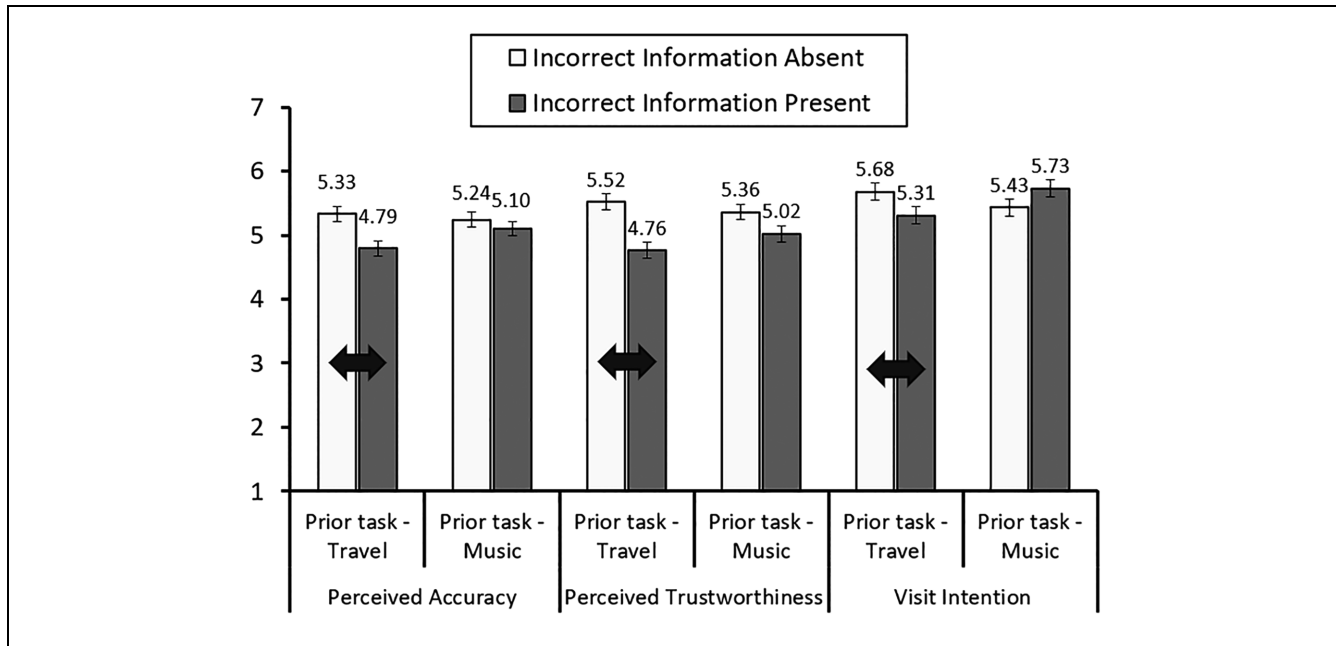


Figure 8. Results of Study 4.

*Error bars represent the standard error. “↔” indicates the significant difference at $\alpha = .05$.

exposed to the mistaken listing of “Robert Schumann (1810-1856)” as a composer who lived before 1800. Participants in the absent condition were not. After reading the information and as the manipulation check, participants were asked to rate ChatGPT’s general accuracy along a 7-point scale (1 = always not at all accurate, 7 = always very accurate).

All participants were then asked to imagine that they were planning a trip to New Zealand’s South Island. They were further informed that they would be using ChatGPT for recommendations about 10 places to visit. All participants were then asked to measure (i) perceived accuracy, (ii) perceived trustworthiness, and (iii) the visit intention on the same scale as Study 1. Participants were then asked to evaluate the realism of the study and to provide their demographic details.

Results and Discussion

The manipulation check was successful in that participants in the incorrect information present condition rated the general accuracy of ChatGPT as lower than those in the incorrect information absent condition ($M_{\text{incorrect information absent}} = 5.03$, $SD = 1.11$ vs. $M_{\text{incorrect information present}} = 3.19$, $SD = 1.71$, $F(1, 331) = 136.16$, $p < .001$, $\eta^2 = 0.291$). The perceived realism was relatively high in that the value was higher than the neutral point ($M = 5.75$, $SD = 1.12$ vs. “4,” $t(334) = 28.59$, $p < .001$).

The research team conducted 2×2 ANOVA for three measurements. First, for visit intentions, two main effects

were not significant (incorrect information: $F(1, 331) = 0.07$, $p = .787$, $\eta^2 < 0.001$; type of information: $F(1, 331) = 0.47$, $p = .509$, $\eta^2 = 0.001$). Importantly, the interaction effect was significant ($F(1, 331) = 6.72$, $p = .010$, $\eta^2 = 0.020$), as is shown in Figure 8. Planned contrast indicated that when the incorrect information was in the same domains, the visit intention was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.68$, $SD = 1.18$ vs. $M_{\text{incorrect information present}} = 5.31$, $SD = 1.24$, $F(1, 331) = 4.06$, $p = .045$, $\eta^2 = 0.012$). However, this pattern disappeared when the incorrect information was in a different domain. Specifically, the visit intention was the opposite pattern, though it failed to reach the significant level ($M_{\text{incorrect information absent}} = 5.43$, $SD = 1.31$ vs. $M_{\text{incorrect information present}} = 5.73$, $SD = 1.03$, $F(1, 331) = 2.72$, $p = .100$, $\eta^2 = 0.008$).

Secondly, the results were very similar to the perceived accuracy of the recommendation. As is shown in Figure 8, the interaction effect was marginally significant ($F(1, 331) = 2.98$, $p = .085$, $\eta^2 = 0.009$). Planned contrast indicated that when the incorrect information was in the same domains, the perceived accuracy was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.33$, $SD = 1.08$ vs. $M_{\text{incorrect information present}} = 4.79$, $SD = 1.15$, $F(1, 331) = 10.76$, $p = .001$, $\eta^2 = 0.031$). However, this pattern disappeared when the incorrect information was in the different domains ($M_{\text{incorrect information absent}} = 5.24$, $SD = 1.02$ vs. $M_{\text{incorrect information present}} = 5.10$, $SD = 0.98$, $F(1, 331) = 0.74$, $p = .392$, $\eta^2 = 0.002$).

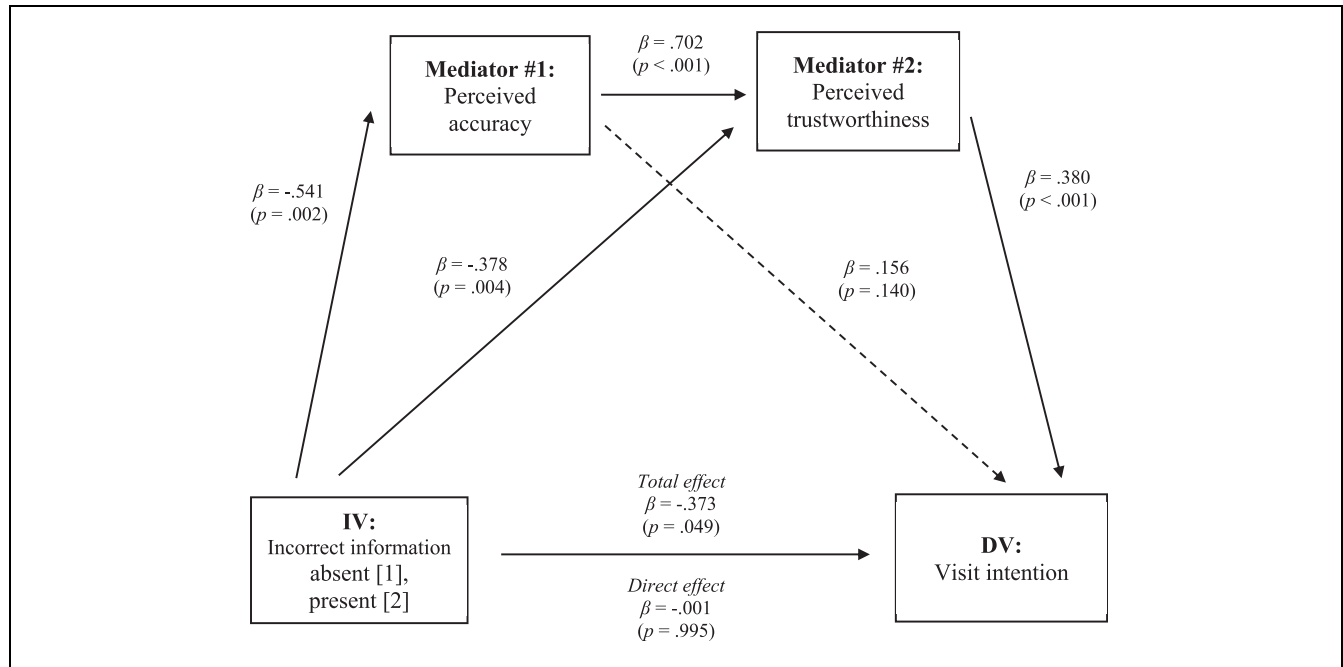


Figure 9. Serial Mediation Results of Study 4 for the Same Domains.

*Indirect effect.

- Incorrect information → Perceived accuracy → Visit intention, 95% CI [-0.240, 0.025].
- Incorrect information → Perceived trustworthiness → Visit intention, 95% CI [-0.299, -0.028].
- Incorrect information → Perceived accuracy → Perceived trustworthiness → Visit intention, 95% CI [-0.292, -0.031].

Thirdly, the results for the perceived trustworthiness of the recommendation were very similar. The interaction effect was marginally significant ($F(1, 331) = 2.80$, $p = .095$, $\eta^2 = 0.008$), as shown in Figure 8. Planned contrast indicated that when the incorrect information was in the same domains, the trustworthiness was higher when incorrect information was absent versus present ($M_{\text{incorrect information absent}} = 5.52$, $SD = 1.06$ vs. $M_{\text{incorrect information present}} = 4.76$, $SD = 1.18$, $F(1, 331) = 18.36$, $p < .001$, $\eta^2 = 0.053$). However, this pattern was much weaker when the incorrect information was in the different domains ($M_{\text{incorrect information absent}} = 5.36$, $SD = 1.16$ vs. $M_{\text{incorrect information present}} = 5.02$, $SD = 1.14$, $F(1, 331) = 3.80$, $p = .052$, $\eta^2 = 0.011$).

Finally, the researchers conducted a serial mediation for two conditions with different types of information. For the same domains, the serial mediation (IV → perceived trustworthiness → recommendation satisfaction → visit intention) with Hayes (2017) macro with 5,000 bootstrapping was significant. For the same domains, the results indicated that the indirect effect of serial mediation was significant (95% CI [-0.292, -0.031]), supporting H2. The detailed results are illustrated in Figure 9. In contrast, for the different domains, the serial mediation (IV → perceived trustworthiness → recommendation satisfaction → visit intention) was not significant (95% CI [-0.204, 0.057]).

Study 5: Testing Boundary Conditions for H5

In Study 5, we examined H5, a boundary condition for the previous hypothesis (H4). Specifically, we anticipated that when people focus on selecting one option in the initial task rather than evaluating the perceived accuracy of the initial information, the negative effect will be reduced, regardless of the relationship between the initial and subsequent tasks.

Method: Participants, Design, and Procedure

The 325 study participants were from the USA ($M_{\text{age}} = 40.12$, $SD = 12.34$; 53.2% female) and were drawn from Amazon Mechanical Turk in exchange for a nominal payment. Participants were randomly assigned to one of 2 (incorrect information: absent vs. present) $\times$ 2 (type of information: similar domains vs. different domains) conditions with a between-subjects design.

The study procedure resembled Study 4, with a few exceptions. The first task was an information search with ChatGPT for either (i) listing 20 destinations of New Zealand's North Island or (ii) 20 classical composers who lived before the year 1800. The mistaken information was 17th on the list, and the presence or absence of the inaccuracy was manipulated in the same way as Study 4. All

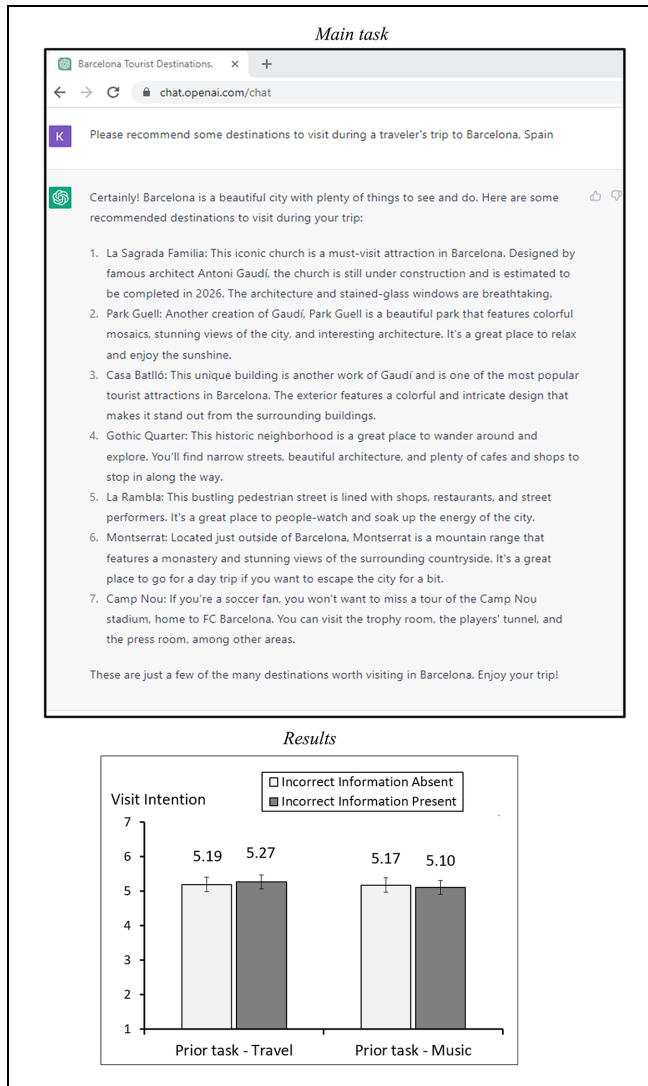


Figure 10. Stimuli and Results of Study 5.

*Error bars represent the standard error.

participants were then asked to choose one option among the first three. They were then asked to imagine that they planned to travel to Barcelona, Spain. They were further informed that they turned to ChatGPT for recommendations about places to visit and were exposed to the seven destinations outlined in Figure 10. After that, all participants were asked to measure visit intentions on the same scale applied in Study 1. Participants were finally asked to evaluate the realism of the study and to provide their demographic details.

Results and Discussion

The perceived realism was relatively high in that the value exceeded the neutral point ($M = 5.57$, $SD = 1.20$ vs. “4,” $t(324) = 23.70$, $p < .001$). Secondly, the choice of initial decision was not significantly different from the incorrect

information manipulation for travel choice ($\chi^2(2) = 0.160$, $p = .923$), though it was marginally significant for the music-related choice ($\chi^2(2)$, $p = .090$).

We conducted a 2×2 ANOVA for visit intentions. The main effect of the domain was insignificant ($F(1, 321) = 0.61$, $p = .435$, $\eta^2 = 0.002$). More importantly, the interaction effect was insignificant ($F(1, 321) = 0.34$, $p = .562$, $\eta^2 = 0.001$), and the main effect of incorrect information was also insignificant ($F(1, 321) = 0.01$, $p = .962$, $\eta^2 < 0.001$). The pattern (Figure 10) was quite different from the Study 4 results.

In sum, the study findings demonstrated the boundary condition for the negative effect of incorrect information in the initial task on the acceptance of the subsequent task. The negative effect was significantly reduced when users were busy selecting one option in the initial task.

General Discussion

Summary of Empirical Studies

This research has explored the effects of incorrect ChatGPT-generated information and of the underlying mechanism in searching for travel information.

Study 1 provided initial evidence of the negative effects of ChatGPT-generated incorrect information on customer visit intentions. Participants demonstrated higher visit intentions, perceived accuracy, and trustworthiness when incorrect information was absent (vs. present). Furthermore, the indirect effect of serial mediation (IV $\rightarrow$ perceived accuracy $\rightarrow$ perceived trustworthiness $\rightarrow$ visit intention) was found to be significant. Studies 2A and 2B provided convergent evidence of the negative effect of ChatGPT's incorrect information when the error was not explicitly mentioned, as well as the robust results obtained from different participants and times.

Study 3 tested the moderating role of the prominence of incorrect information. The results demonstrated that in the absence of incorrect information, the intention to visit was similar, regardless of the prominence of the incorrect information (in this case, position). On the other hand, the visit intention was much lower when incorrect information had a prominent (vs. a less prominent) position.

Study 4 investigated the moderating role of the prominence of incorrect information on the subsequent decision. When the incorrect information was in the same domains, visit intentions, perceived accuracy, and perceived trustworthiness were higher when incorrect information was absent (vs. present). This pattern, however, disappeared or weakened when the information was in the different domains. In addition, the indirect effect of serial mediation (IV $\rightarrow$ perceived trustworthiness $\rightarrow$ recommendation satisfaction $\rightarrow$ visit intention) was found to be significant for the same domains, but not for the different domains. Study 5 examined a boundary condition for

the negative effect of the incorrect information of the initial task on the acceptance of the subsequent task. The findings demonstrated that the negative effect was significantly reduced when participants focused on the initial task.

Theoretical Implications

This research makes several theoretical contributions. First, the study has extended the literature on AI-based chatbots by demonstrating the potentially negative impact of offering incorrect information through such technologies and by exploring traveler acceptance of its recommendations. The study findings highlight the potential limitations of such technologies in providing accurate travel-related information. While previous investigations focused on the initial acceptance phase of AI technologies in tourism contexts (Gursoy et al., 2019; Sun et al., 2019), the current study sheds light on the post-adoption phase of AI-based chatbots in tourism. Scholars have previously noted the importance of understanding tourists' post-adoption behaviors for AI-based technologies (Shi et al., 2021; Xiang et al., 2015). The current study addresses the need for such contributions.

Second, by providing initial evidence of the potentially negative impacts of ChatGPT on traveler attitudes and behaviors, the study findings shed light on the need for a more conscious approach toward technology and raise valid concerns regarding technology developments and their use in tourism and beyond. There is a significant difference between human learning and AI, which are based on artificial neural networks and deep learning. It can lack a comprehensive understanding of context and of ethical considerations. It is recognized that these distinctions contribute to informed decisions about the use of AI and its limitations, especially in areas where human values, ethics, and creativity are essential. The recent emergence of AI like WormGPT or DarkBERT underscores the importance of approaching new technologies with caution, especially when we cannot discern the learning processes or underlying value systems behind the solutions offered by AI. Relatedly, the mechanistic worldview shared by AI and neoliberal economics exemplified by the German Smart City Charter (2017) poses a threat to healthy democracies. The increasing use of behavior-related data may replace traditional democratic methods like polls and voting, potentially eroding democratic principles and raising ethical and social concerns such as discrimination. Moreover, it is essential to note that critics highlight shared limitations of AI and neoliberalism, including the potential to overlook ethical and moral concerns, reduce human cognition and intelligence, oversimplify human behavior, and display inflexibility to change (M. Fuchs, 2022; Mikalef et al., 2022).

A more comprehensive and holistic perspective, coupled with a better understanding of inherent weaknesses, has potential to mitigate the negative effects of the mechanistic view associated with AI and neoliberalism, leading to the development of a more humane AI and healthier economic policy. This aligns with a previous scholarly emphasis on the need to examine unchecked technological development critically. For instance, Gretzel et al. (2020) argue for limiting and democratizing technology development in tourism to promote equity and sustainability. C. Fuchs (2009) questioned technological determinism, advocating for constraints on technological advancement to foster innovative decision-making. Tribe and Mkono (2017) caution against excessive reliance on technology in tourism, which can lead to alienation rather than genuine human connections. While chatbots and generative AIs offer efficiency and convenience, their limitations must be considered carefully for an alignment with human interests and values.

Third, based on the accessibility-diagnostics framework (Herr et al., 1991; Menon et al., 1995), this study has contributed a conceptual relationship between the domains of incorrect information and information salience. In so doing, the researchers investigated the boundary conditions that moderate the negative impact of incorrect ChatGPT-generated information. The study results have identified the prominence of incorrect information as the boundary condition. As previous researchers have demonstrated, the ways by which consumers form their attitudes toward misinformation are influenced by various factors, including cross-cultural differences (Borges-Tiago et al., 2020), product types (Chang et al., 2020), and religiosity (Wisker, 2021). The current research extends understanding of how ChatGPT can be used effectively when providing travel recommendations.

Fourth, this study has examined the impact of prior exposure to incorrect information on subsequent tasks. The results provide insights into the effects of chatbot recommendations on user trust and reliance. The travel decision-making process is complex when planning trips, and travelers face many uncertainties (Mcknight et al., 2011; S. Park & Tussyadiah, 2017). Furthermore, prospective travelers are dependent on the availability of information; hence misinformation may change how tourists collect and process information (Williams et al., 2022). Accuracy and trustworthiness can evidently transform and influence behavioral attitudes and intentions toward AI-based products and services (Chang et al., 2020; Cheng & Chen, 2021; Shi et al., 2021).

Practical Implications

The preceding investigation has several implications for practitioners. First, although AI integration promises efficiency and competence, our results indicate that AI-based

travel planning tools may have long-lasting negative effects on customer attitudes and behaviors when they recognize the inaccuracy of AI-provided information. Thus, practitioners must work on mitigating errors using various tools, such as multi-step verifications at different stages. Additionally, incentivizing customer feedback on errors can help correct AI errors promptly. It is noteworthy that the current research revealed upfront that some ChatGPT-provided information is wrong. However, these errors may not be evident to most consumers, unless they are highly knowledgeable about travel destinations. Thus, more careful attention should be paid to mitigating errors, especially in recommending well-known and popular destinations.

Second, our results have shown that perceived trustworthiness mediates the relationship between inaccurate information and a lower willingness to adopt AI recommendations. Therefore, in addition to mitigating errors, practitioners can enhance cognitive and emotional trust in AI (Glikson & Woolley, 2020). Recent achievements, such as ChatGPT passing the US bar exam or the Turing test (Koetsier, 2023), have already established people's confidence and trust in AI tools. Practitioners may exert leverage by showing that ChatGPT powers their AI. Moreover, ChatGPT's eloquence, choice of words, and quality of answers make the content both convincing and persuasive, thereby building emotional credibility. Thus, practitioners should consider how to keep ChatGPT's human-like conversational style when recommending personalized travel plans.

Third, our results have shown that the negative impacts of inaccurate information are not carried over into other domains. This suggests that practitioners should minimize the spillover effect of AI-based recommendation errors by carefully planning their website user interface/ user experience designs.

In aggregate, this research has provided thorough insights into the potentially negative impacts of the provision of incorrect information by AI-based chatbots on traveler acceptance of its recommendations. Since tourists rely heavily on the accuracy and credibility of information when evaluating and accepting recommendations (C. M. K. Cheung et al., 2008; D. H. Park et al., 2007), our study findings highlight the potential limitations of ChatGPT in providing accurate travel information. This study also deepens our knowledge about the factors influencing users' trust and reliance on such technologies in the context of travel recommendations.

Limitations and Opportunities for Future Research

This study has several limitations and these suggest some future research directions. First, we deployed an experimental design by presenting respondents with non-explicit incorrect information in Studies 2A and 2B. For instance,

our preliminary results indicated that awareness could be a significant moderating factor, suggesting that the incorrect information about Paris and London was noticed more by UK participants than by their US counterparts. This resulted in a stronger effect on visit intentions, perceived accuracy, and trustworthiness. However, as awareness and the ability to identify fake news and misinformation may vary among individuals (Borges-Tiago et al., 2020), further investigation is needed to control for such potential differences amongst consumers.

Second, our current empirical studies have relied primarily on behavioral intention (e.g., visit intention). To enhance ecological validity and external validity, it would be useful for future studies to extend the different measurements, including preference² or actual behavioral measurements.

Third, the knowledge cutoff date of ChatGPT for this study is September 2021. Information about ChatGPT will likely evolve quickly over time, allowing for potentially different findings in the future. However, this study has provided a snapshot of ChatGPT's impact and offers valuable insights within the aforementioned timeframe. Future replication studies could potentially build upon the current research. In addition, we suggest that future studies should examine and compare the advancements and differences or changes in ChatGPT to provide more up-to-date insights.

Fourth, the current researchers also call for future investigations to explore the ethical use and application of information communication technologies (ICTs). As the utilization of ICTs significantly impacts firms' value creation and strategic development (M. Fuchs & Sigala, 2021), an increasing body of research has explored the strategic utilization and benefits of ICTs in various industries, including in travel and tourism (Buhalis et al., 2011; M. Fuchs & Sigala, 2021; Hughes & Moscardo, 2019). Specifically, the integration of ChatGPT is expected to bring profound changes to the hospitality and tourism sector, altering how consumers seek information and make decisions and how businesses deliver services and experiences (Dwivedi et al., 2023). However, there is evidently a potential downside of ICTs as they can generate and disseminate false information, which may not only mislead customers but also lead to social issues such as political and social polarization. The application of artificial intelligence through channels such as ChatGPT raises ethical concerns, necessitating a careful acknowledgment and consideration of the risks of misinformation and other ethical dilemmas (Thompson, 2023). For example, as proposed by Thompson (2023), Buddhist philosophy, emphasizes the importance of understanding the ethical implications of technological advances such as ChatGPT. This paper highlighted the need to approach modern complexities responsibly and compassionately, keeping in

mind the potential impact of such technologies on society and individuals. In conclusion, the increasing influence of ICTs on business and industries such as travel and tourism has sparked a need for vigilance regarding their ethical implications. While ChatGPT holds the promise of transformative benefits, its potential to spread false information calls for responsible and compassionate consideration of its application (Thompson, 2023). Acknowledging and addressing these ethical concerns can guide us in navigating the modern landscape in a way that promotes collective well-being. In extending the current research, it will be imperative to direct more attention toward the ethical use and application of ICTs.

To address concerns arising from the mechanistic worldview of AI and neoliberal economics, and as exemplified by the German Smart City Charter (2017), it is essential to balance the efficiency and insights provided by technology and the preservation of democratic values and processes. Importantly, researchers should endeavor to provide a more nuanced understanding of the challenges and opportunities presented by the intersection between AI, economics, and democratic governance. Future researchers should adopt a comprehensive approach when assessing the actual impact of AI and neoliberal economic principles on democratic processes and societal well-being, offering practical recommendations to safeguard human and social values in an AI-driven world.

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All authors contributed equally to this research.

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Notes

1. Therefore, all participants were exposed to the same 20 options. Since New Zealand destination is less familiar

to US participants ($M = 2.10$ out of 1–7 scale of familiarity [1 = not at all familiar, 7 = highly familiar]), the incorrect information is not distinctive in the absent condition.

2. In order to test the impact of the presence (vs. absence) of incorrect information on the preference for the information source, we conducted a post-hoc test with 268 US MTurkers ($M_{\text{age}} = 41.32$, $SD = 12.52$; 56.0% female) using a one-way between-subjects design (incorrect information: absent vs. present). First, participants were exposed to the section on “Information search with ChatGPT,” which included the similar material as Study 2A (including incorrect information or its absence regarding the recommendation of a trip to Paris, France). Afterward, all participants were asked to imagine that they were planning to travel to Kyoto and were instructed to choose one information source from three alternatives: ChatGPT, an experienced friend, or an online travel agent (OTA). This choice was based on the materials from Study 4 by Kim et al. (2023a). The results indicated that the preference for ChatGPT was higher when incorrect information was absent (vs. present) ($M_{\text{absent}} = 21.5\%$ [29/135] vs. $M_{\text{present}} = 12.0\%$ [16/133], $\chi^2(1)$, $p = .038$). Additionally, the result of a binary logistic regression analysis indicated that this effect was robust, irrespective of participants’ usage of ChatGPT, destination familiarity, travel experience, age, and gender. Detailed results are available from the authors.

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